

Tisaben Dineshkumar Patel

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GitHub: github.com/TisaDR | Portfolio: <https://my-portfolio-ebon-gamma-72.vercel.app/>

SUMMARY

Full-stack developer with a strong AI/ML focus, currently completing a Graduate Certificate in IT Solutions. Built and deployed an end-to-end evaluation harness for LLM-generated healthcare answers, with custom metrics, a curated test dataset, and versioned run tracking. Experienced with the Claude API, prompt engineering, and testing autonomous AI agents. Comfortable across frontend, backend, and data, and interested in healthcare and virtual care.

SKILLS

Languages & Web: Python, JavaScript, HTML, CSS, SQL, REST APIs, FastAPI, Chart.js

AI / ML & Evaluation: Anthropic Claude API, OpenAI API, Groq API, prompt engineering, agentic AI systems, LLM eval metrics, scikit-learn, TF-IDF

Data: PostgreSQL, data analysis, dataset curation & versioning, Jupyter, data visualization

Tools: Git/GitHub, VS Code, Postman, Render, Neon, WordPress; Python data-visualization dashboards

PROJECTS

LLM Evaluation Harness for Healthcare Q&A

Personal Project

- Built a full-stack evaluation harness (FastAPI, PostgreSQL on Neon, HTML/JS) that automatically scores LLM-generated healthcare answers across four custom metrics: accuracy, safety, relevance, and length.
- Designed a safety metric that flags overconfident or unsafe medical language and missing emergency guidance, catching failure modes that keyword-accuracy checks miss.
- Curated a 50-question test dataset (symptoms, medication, mental health, emergencies, pediatrics) with expert-authored ground-truth keywords; combined metrics into a weighted overall score.
- Versioned the schema (eval_runs → evaluations) so every batch of scored answers is tied to a run, enabling comparison of model and prompt changes over time; added a Chart.js trend dashboard and live Q&A testing.
- Integrated live Groq API calls (gpt-oss-120b) with a safety-focused system prompt. Deployed on Render and resolved dependency, SSL, model-deprecation, and leaked-secret issues independently.

Live demo: <https://healthcare-eval.onrender.com/> | Code: [\[add repo URL\]https://github.com/TisaDR/healthcare-eval](https://github.com/TisaDR/healthcare-eval)

Agentic AI Testing (Capstone)

Academic, 2024 – 2025

- Designed and ran a test suite for autonomous AI agents, evaluating reliability, reasoning accuracy, and task completion across multi-step workflows.
- Compared agent behavior on structured vs. unstructured inputs and delivered evaluation reports with metrics, findings, and recommendations.

Claude API Integration & Prompt Engineering

Personal Project, 2024 – 2025

- Built a VS Code–integrated tool calling the Anthropic Claude API, engineering prompts to control output structure; implemented authentication, request handling, and response parsing.

Shopify AI Assistant

Personal Project, 2024 – 2025

- Built an AI assistant on a Shopify storefront to handle product queries and recommendations; explored how structured content affects discoverability.

RESEARCH & EXPERIENCE

Machine Learning Researcher | Mental Health Disorder Prediction

Charusat University

- Compared Logistic Regression, SVM, and Random Forest to classify 4 mental health disorders from 17 behavioral symptoms, reaching 95.33% accuracy.
- Owned the full ML pipeline (cleaning, encoding, feature selection, training, evaluation); published as a fellowship paper with a 27-source literature review: doi.org/10.5281/zenodo.20405049

Mobile Application Developer -Fellowship| Charusat University

2023 – 2024

- Developed a mobile app that captures patient input in digital form and built visuals to analyze patient data.

EDUCATION

Graduate Certificate, Information Technology Solutions | Humber College

Jan 2025 – Aug 2026

Coursework: deep learning, machine learning, data analysis, web development

B.Sc., Information Technology | Charusat University

Sep 2021 – May 2024